

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc Physics Examination April-May 2018

PS958 Semiconductor Devices

Date: 30-04-2018, Day: Monday Time: 10.00 AM To 10.20 AM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q-1 Choose the correct answer for the following questions

[10]

1. The width of the space charge region increases as the applied _____ increases.
(a) Forward voltage
(b) Reverse voltage
(c) Temperature
(d) Majority carriers
2. Current flow in a semiconductor device depends on the phenomenon of
(a) Diffusion
(b) Drift
(c) Recombination
(d) All of the above
3. A MOSFET is a _____ controlled _____ carrier device.
(a) voltage, majority
(b) voltage, minority
(c) current, minority
(d) current, majority
4. The ideal reverse-saturation current density of the Schottky barrier junction is orders of magnitude ----- that of the ideal pn junction diode.
(a) larger than
(b) smaller than
(c) equal to

- (d) both are not related
5. The depletion type of MOSFET can operate in the:
 - (a) depletion mode only
 - (b) enhancement mode only
 - (c) in the depletion mode and the enhancement mode
 - (d) none of the above
 6. A major disadvantage of MOSFETs is:
 - (a) its high input impedance
 - (b) that it is a voltage operated device
 - (c) that it is sensitive to electrostatic discharges
 - (d) none of the above
 7. Which of the following statement/s of a Schottky diode is true?
 - (a) In a Schottky diode there is essentially no recombination in the depletion region.
 - (b) The ideality factor is very close to unity.
 - (c) The Schottky barrier is a majority carrier device.
 - (d) all the above
 8. An n-semiconductor as a whole is:
 - (a) Negatively charged
 - (b) Positively charged
 - (c) Electrically neutral
 - (d) Negatively or positively charged depending on doping.
 9. The forward bias current in a typical Schottky barrier is due to what physical mechanism?
 - (a) Drift
 - (b) Diffusion
 - (c) Recombination
 - (d) Thermionic emission
 10. To make a good ohmic contact to a semiconductor, what should be done?
 - (a) Choose a metal with a high Schottky barrier height.
 - (b) Use a lightly doped semiconductor
 - (c) Introduce defects into the semiconductor to lower its lifetime
 - (d) Dope the semiconductor very heavily

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PS958 Semiconductor Devices

Date: 30-04-2018, Day: Monday Time: 10.20 AM To 12.00 Noon Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

- Q - II Answer the following questions as directed** **10**
1. (a) Forward current density in a diode is inversely proportional to the life time of carriers. State: True/False 2
(b) In an unbiased p-n junction, the junction current at equilibrium is diffusion of majority carriers only. State: True/False
 2. Consider a p-type Si sample with 10^{16} donors cm^{-3} . The two ends of the sample are labeled A and B. The work function of silicon, Si is 4.15 eV and the work function of two potential metals for contacts at A and B are Li ($\Phi_m = 2.5\text{eV}$) and Au ($\Phi_m = 5\text{eV}$). 2
(a) Ideally, which metals will result in a Schottky contact?
(b) Ideally, which metals will result in an Ohmic contact?
Explain your results.
 3. What is IGFET, MESFET, MODFET and HFET? 2
 4. Calculate the value of V_{bi} and Φ_m in an ideal metal-Si Schottky barrier contact. Assume the barrier height is 0.8 eV, $N_C = 2.86 \times 10^{19} \text{ cm}^{-3}$, $N_D = 1.5 \times 10^{16} \text{ cm}^{-3}$ and $q\chi = 4.01 \text{ eV}$. 2
 5. What are heterojunctions? What are uses of such junctions in the device technology? 2

SECTION - II

- Q-III Answer the following questions as directed (any five)** **15**
1. What is an ohmic contact? What are the significance of good ohmic contacts? Explain the methods of achieving good ohmic contacts. 3
 2. What are the main constructional differences between a MOSFET and a JFET. What effect do they have on the current conduction mechanism of a MOSFET? 3
 3. Discuss the formation of Schottky junction between a metal and n-type semiconductor. Show the energy band diagram under equilibrium. 3
 4. Copper is deposited on a carefully prepared n-type silicon substrate to form an ideal Schottky diode. Taking, $\Phi_m = 4.65 \text{ eV}$, electron affinity = 4.01 eV, $N_D = 3 \times 10^{16} / \text{cm}^3$ and $T = 300 \text{ K}$ calculate the barrier height, the built-in potential and the depletion layer width at a zero bias condition. 3
 5. Write significant differences between p-n junction diode and a Schottky barrier diode. Compare the current-voltage characteristics of Schottky barrier diode and p-n junction diode. 3
 6. Draw and explain the depletion-mode and the enhancement-mode operation of D MOSFET 3